

Educational ARES User Guide



1 INSTALLATION AND SETUP

1.1 DOWNLOAD ARES

Download and place ARES on your system. These files can be placed anywhere on your computer.

1.2 INSTALL YOUR CERTIFICATE

For ARES to have secure communication between its user interface and service, you will need to install the service certificate on your machine. For your convenience this has been provided in the form of a script. In your ARES install you will need to run your “InstallCertificates.ps1” file.

1.3 UPDATE YOUR DATA PATH

ARES uses a user specified path to save information. This includes campaign templates, experiment output and your SQLite database. By default, this path is set to “C:\ARES\Data”. If you would like to change this, navigate into the ARES installation folder. Under the service folder, open the file named “appsettings.json”. The ARES data path is defined in the highlighted line in the image below. Update the “C:\ARES\Data” to whatever absolute path you would like your data to be saved on, then save and close the file.

```
1 {
2   "TokensConfig": {
3     "Issuer": "localhost",
4     "Audience": "localhost",
5     "Key": "Some private key that is definitely over 128 bits in length"
6   },
7   "ConnectionStrings": {
8     "CoreDatabase": "Data Source=educationalares.db"
9   },
10  "Logging": {
11    "LogLevel": {
12      "Default": "Information",
13      "Microsoft": "Warning",
14      "Microsoft.Hosting.Lifetime": "Information"
15    }
16  },
17  "AllowedHosts": "*",
18  "Kestrel": {
19    "EndpointDefaults": {
20      "Protocols": "Http2"
21    },
22    "Endpoints": {
23      "Http": { "Url": "http://localhost:5000" },
24      "Https": { "Url": "https://localhost:5001" }
25    },
26    "Certificates": {
27      "Default": {
28        "Path": "FC2ServiceCert.pfx",
29        "Password": "SecurePassword"
30      }
31    }
32  },
33  "CertificateSettings": {
34    "Path": "FC2ServiceCert.pfx",
35    "Password": "SecurePassword"
36  },
37  "AresDataPath": "C:\\ARES\\Data",
38  "DemoDevicePort": 7038,
39  "DemoPlannerPort": 7069,
40  "DemoAnalyzerPort": 7086,
41  "PrintPlannerPort": 8006,
42  "PrintAnalyzerPort": 8007
43 }
44
```

1.4 STARTING ARES

Provided with your copy of ARES is a batch file called “StartARES.bat”. This will handle starting both the user interface and the service. You can make a shortcut to this file by right clicking on it and selecting “Create Shortcut”. By default, you can access the ARES UI on localhost:7084. Open any web browser and navigate to this address. If you can see the ARES UI, then the program is running properly.

2 NAVIGATING ARES OS

2.1 THE DASHBOARD

The dashboard tab of ARES is the home page of the program. Here you will have access to the interface of any device that you create. These interfaces offer a manual option for interacting with devices. For information regarding specific device interfaces, please visit their respective sections in the guide.

2.2 AUTOMATION

The automation tab consists of two important pages, the campaign designer and the execution page.

2.2.1 The Campaign Designer

The campaign designer is responsible for creating, updating, and deleting campaigns. These campaign templates will provide instructions for ARES to execute during autonomous experimentation. For more information on building campaigns, please see the campaign building section.

2.2.2 The Execution Page

The execution page is responsible for starting campaigns. Here you can pick from a list of your defined campaign templates, set the stop conditions for the campaign and execute, pause or stop your campaign.

2.3 EVENT HISTORY PAGE

The event history page is responsible for displaying a list of all notifications you’ve received from ARES. These notifications will indicate important events, such as a device being added, a campaign being started, or an error occurring during the software’s operation.

2.4 THE SETTINGS MENU

The settings menu is responsible for the management of various ARES assets. These include devices, planners and analyzers, all of which have their own dedicated tabs. All these tabs and their use will be explained in their respective sections.

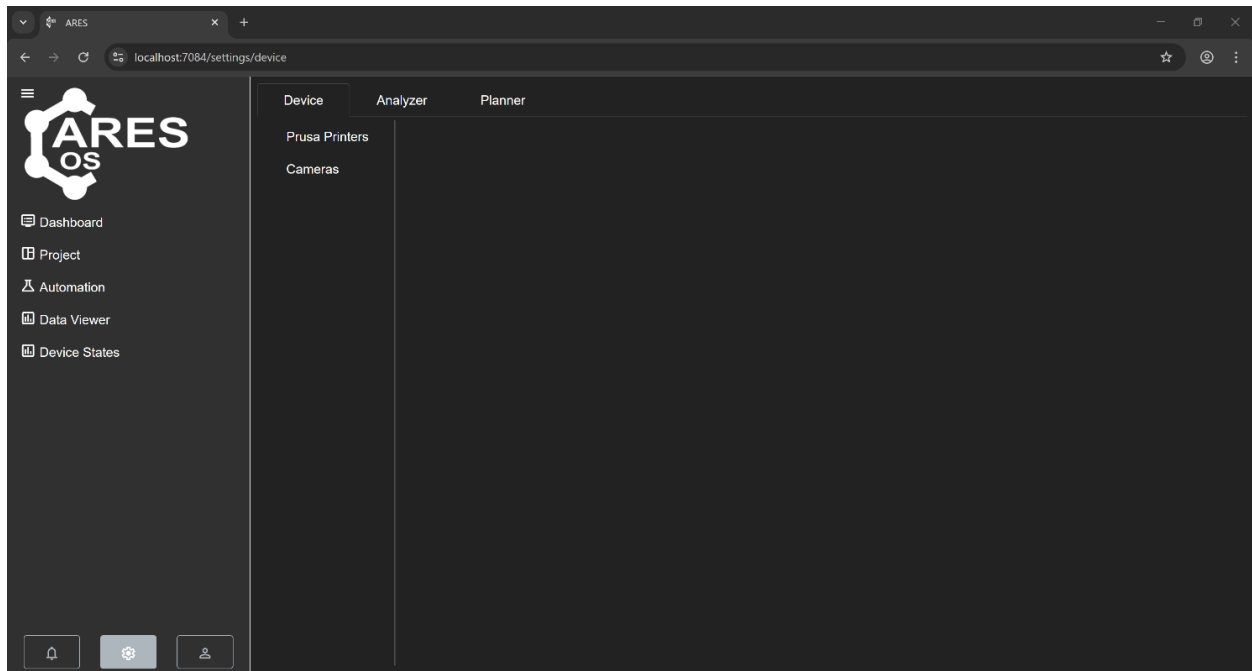
2.5 DOCUMENTATION

The documentation tab is a copy of the information you see here, added to the ARES software for convenient viewing anywhere you can access the ARES webpage.

3 DEVICES

3.1 ACCESSING THE DEVICE SETTINGS MENU

In operating the ARES system, you will utilize two classes of devices. The Prusa MK4S 3D Printer, and a Camera. By default, ARES OS will not have any devices loaded. Instead, you will need to create these device instances in the settings menu. This menu can be found under the gear icon in the bottom left corner. Once you have entered the settings page you will by default be opened to the “Device” tab. Here you will see your two device categories, the Prusa Printer and the Camera.



3.2 PRUSA MK4S PRINTER

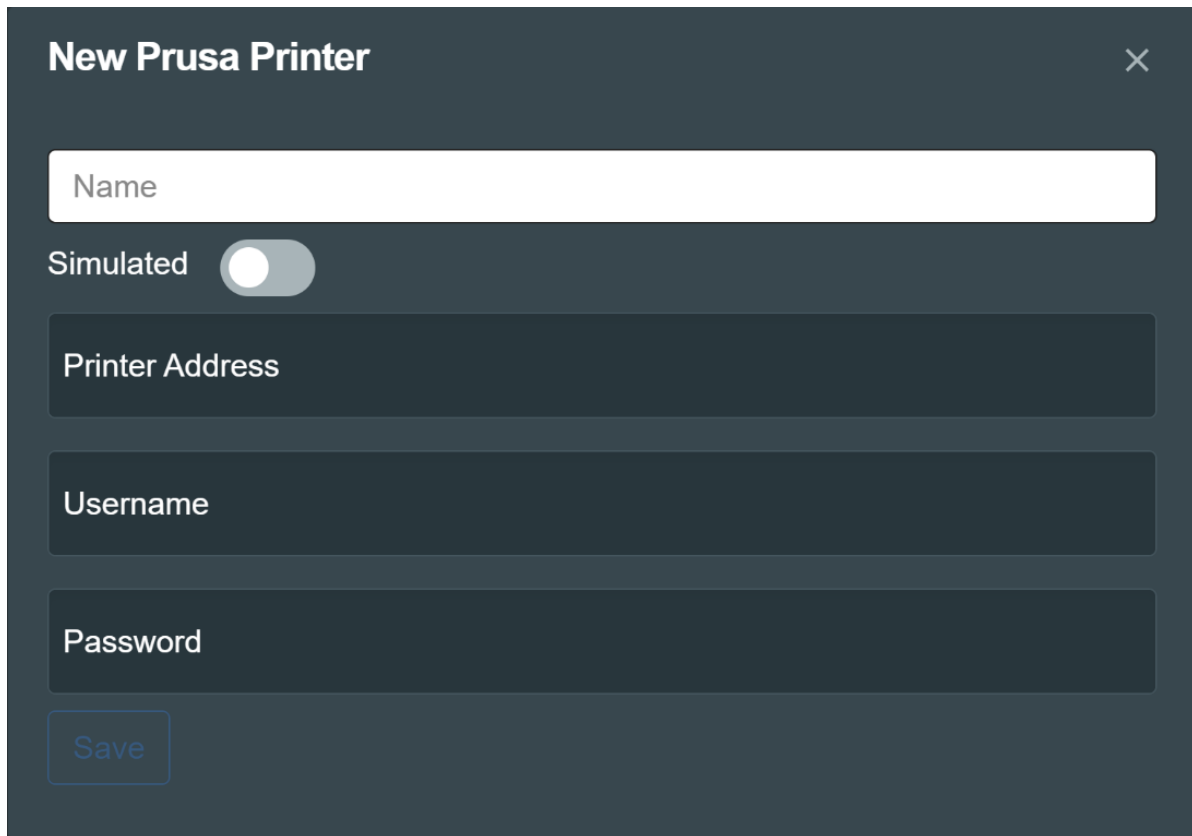
3.2.1 Creating the Printer

In the settings menu, click on “Prusa Printers”. In this menu you will find a button on the far-right side of your screen with a “plus” icon. Clicking on this will open the dialogue screen to add a new instance of the Prusa Printer. There are four required pieces of information to create a printer; a name, address, username and password.

- Name – The friendly device name used throughout ARES.

- Printer Address – The IP address of the printer, this should be formatted as <http://0.0.0.0>, replacing 0.0.0.0 with the printers IP address. This can be found in your printer’s settings menu by going to: Settings → Network → Network Status → IPv4 Address.
- Username – The username used to access your printer. This can be found in your printer’s settings menu by going to: Settings → Network → PrusaLink → User.
- Password – The password used to access your printer. This can be found in your printer’s settings menu by going to: Settings → Network → PrusaLink → Password.

It is important that your address, username and password are correct. Without this information, ARES won’t be able to connect with your printer. With this information is entered, press save. You will see a notification in the top right. It will display green if the device was added successfully and red if it was not. Congratulations, you have successfully added your printer!



New Prusa Printer ×

Name

Simulated ☐

Printer Address

Username

Password

Save

3.2.2 Using the Printer U.I.

The printer’s user interface displays key device information and offers some basic device control. The interface will display “Printer Connected” or “Printer Disconnected” based on whether ARES is successfully communicating with the device. If communications are active with the printer, a real time reading of both the nozzle and bed temperatures will be displayed in the UI as well.

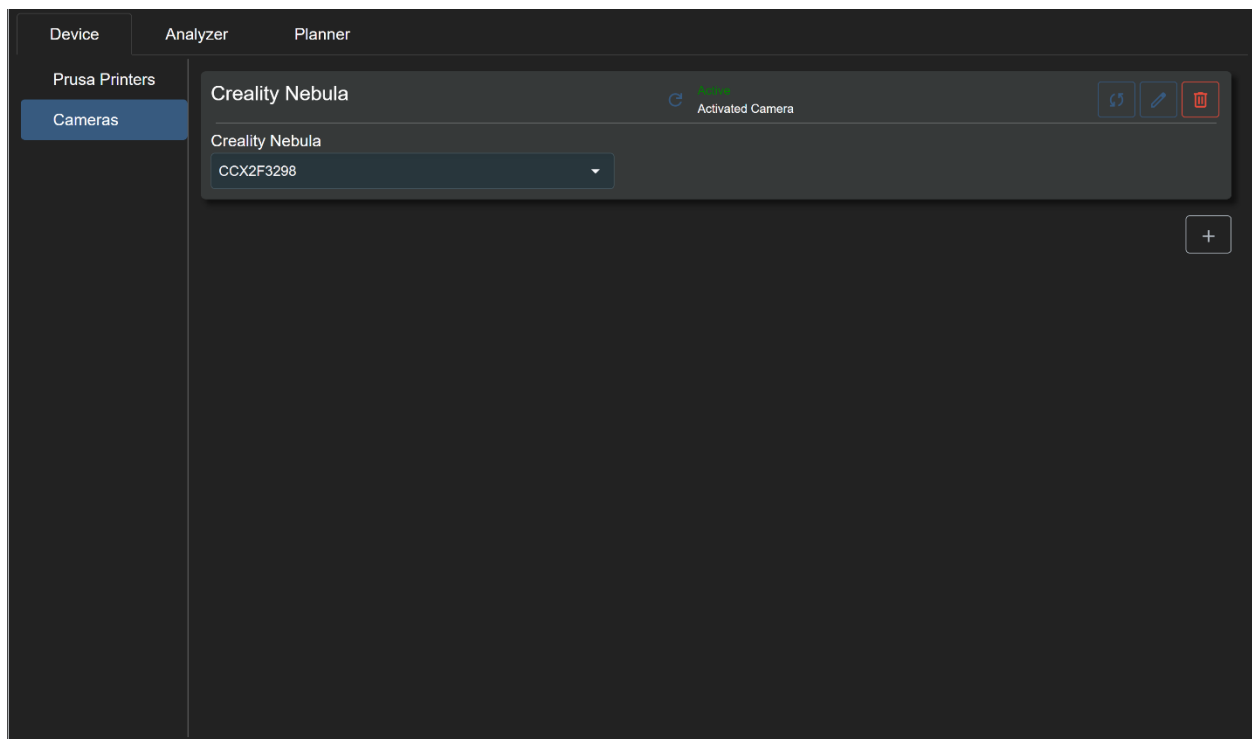
The user interface offers two commands that can be sent to the printer. The first is a simple movement command, with an optional dwell time. The text box default shows the accepted format

of the input; the x, y, z and dwell time values separated by commas. For example, if you wanted the printer to move to the coordinates 100, 100, 100 and wait there for ten seconds, your input would be 100,100,100,10. Once you've inserted your desired values, press the blue play button to execute the command. The user interface also allows the homing command to be sent to your printer by pressing the home button.

3.3 CAMERA

3.3.1 Creating the Camera

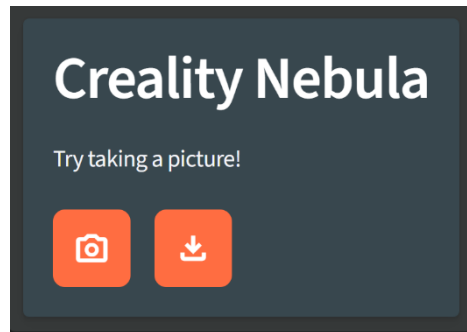
In the settings menu, click on "Cameras". To add a new camera, click the plus button on the far-right side of your screen. Clicking this will open the dialogue screen to add a new instance of a camera. Relative to the printer, the camera is very simple and only requires a name to be created. After entering a name, press save. Once your camera has been created, you will see its entry in your list of camera devices as pictured below. From here you can select your input source. Any camera that is detected by windows as a webcam will appear on this list and can be used as your source. Once you have selected your source from the drop-down menu, your camera is ready to use!



3.3.2 Using the Camera U.I.

The camera's user interface gives the user the ability to take test pictures, as well as download those pictures directly from the dashboard. When no picture has been taken, the interface will display "Try taking a picture!" in place of any image. To take a test picture, press the orange camera button. If the picture is taken successfully, the image will be displayed in the middle of the

interface. Should you wish to download this image, you can do so by simply pressing the orange download button on the interface. These images will be downloaded as PNGs.



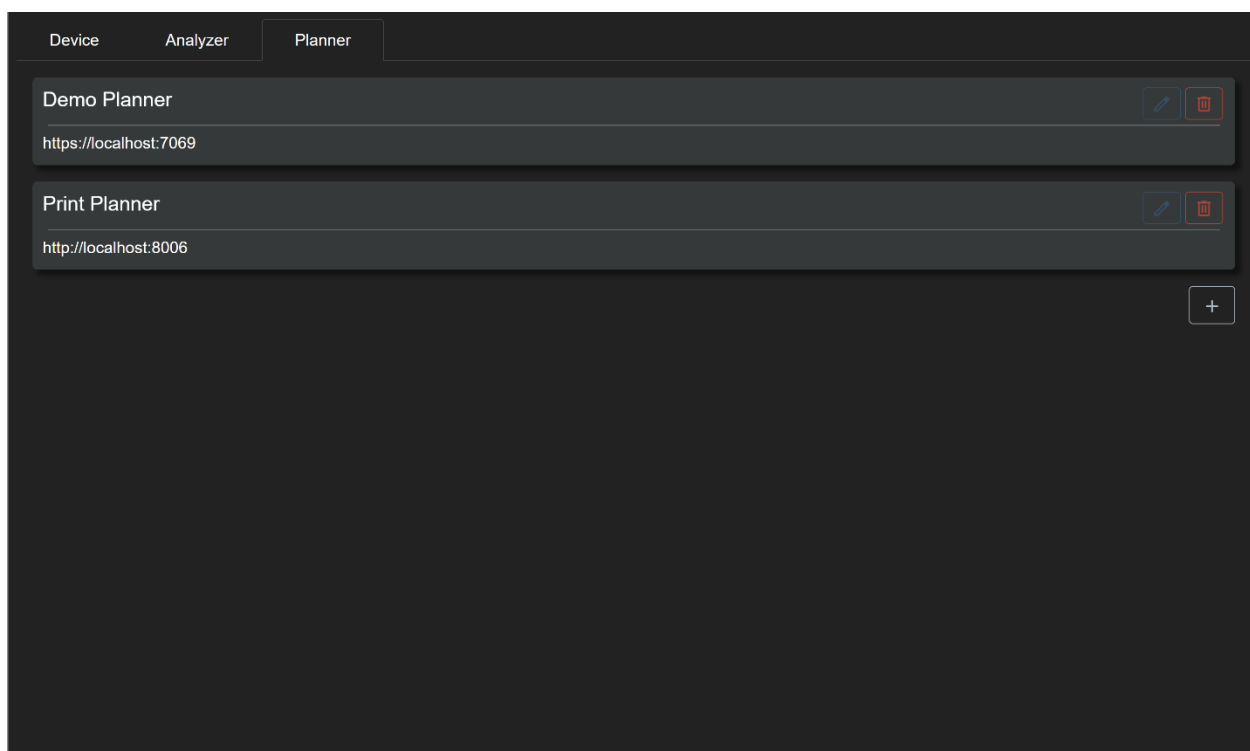
4 PLANNERS AND ANALYZERS

4.1 PLANNERS

Planning is one of the three key components to autonomous experimentation and is responsible for making intelligent decisions about parameter values in experiments. ARES allows for the creation and use of custom planners that communicate over a computer's local network. Follow along to learn how to create your own custom planner to use for your autonomous experiments.

4.1.1 ARES Planner Settings

The settings page in ARES contains a dedicated tab for planners. In this tab a list of available planners is displayed. By default, ARES will have two planners, the demo planner and the print planner. Neither of these can be edited directly from this menu, as they are protected implementations not designed to be changed. For your convenience, the print planner is created by default for communications with your custom-built planner.



4.1.2 Adding New Planners

While ARES by default contains a planner for you to use for your custom planning logic, it also has the capability of creating additional planners. To create a new planner, navigate to the planner tab in the settings menu and click the plus button underneath your last planner entry.

Creating a new planner requires two pieces of information:

- Planner Name – The friendly name ARES uses to refer to the planner.
- Address – The address ARES uses for communications with the planner. By default <http://localhost> will be populated, and in most cases the user will simply add a port number (e.g. <http://localhost:9000> for port 9000).

It's important to give your planner a unique address, otherwise the communication services won't be able to establish their connections. Once these fields are populated, press save. ARES will notify you if an error occurs.

4.1.3 Creating a Custom Python Planner

To support your creation of a custom planner, a python package can be installed that handles the backend work for creating a planner. To install this package, you will need to utilize pip. Run the command “pip install PythonAres” to begin the installation process. The ARES Python package, as well as its dependencies, should be installed.

The key to building your own ARES planner in Python is understanding the support message structure behind it. Your planner will communicate with ARES using the request and response messages defined in the PythonAres library. To build your custom planner, you will need to know the structure of these messages. For a detailed layout of the plan request message, see section 7.

Below is a basic example of creating a new planner that uses port 8006. It's important that the port you use in your Python script matches the port of the “PrintPlanner” in ARES. A basic template for creating one of these planners is included in your ARES installation.

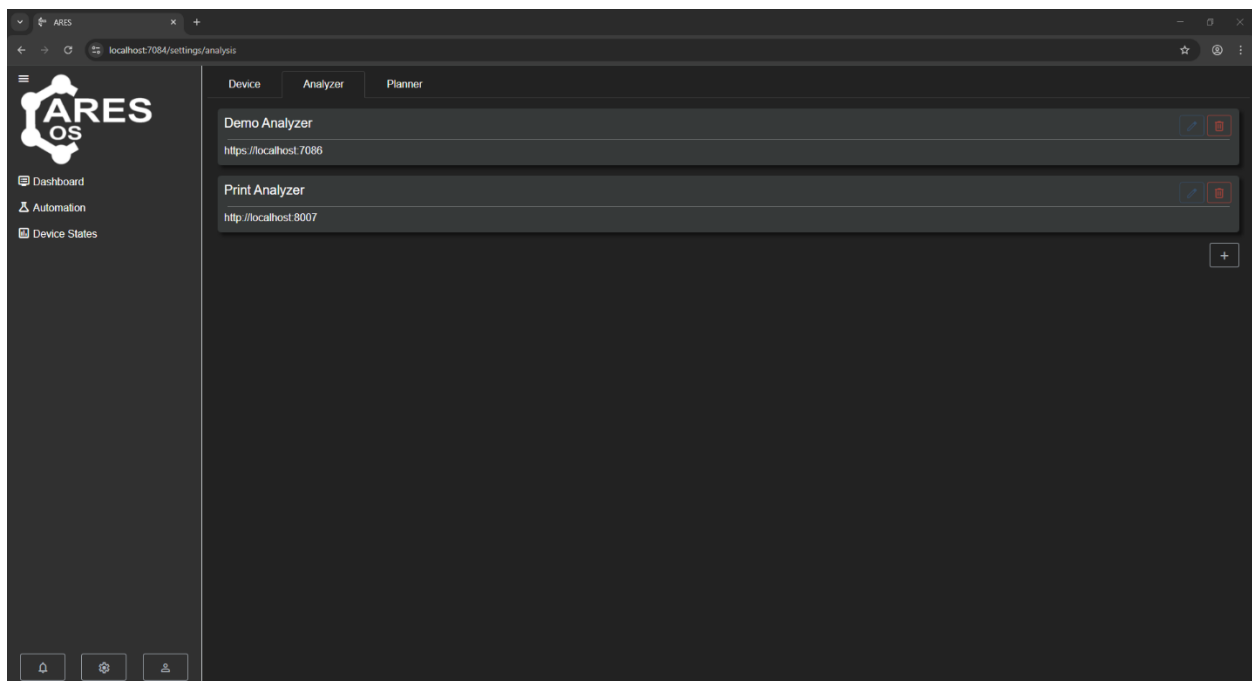
```
1  from PythonAres import PrintPlanner
2  from PythonAres import PrintPlanResponse
3
4  import random
5
6  def GetRandomBedTemp():
7      temperatures = [61, 61, 61, 61, 62, 62, 62, 63, 63, 63]
8      selection = random.randint(0, 9)
9      value = temperatures[selection]
10     print(f"Bed Temp value is being assigned at {value}")
11     return value
12
13     def GetRandomNozzleTemp():
14         temperatures = [225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235]
15         selection = random.randint(0, 10)
16         value = temperatures[selection]
17         print(f"Nozzle Temp value is being assigned at {value}")
18         return value
19
20     def HandlePlanning(request, context):
21         print("ARES Planning Request Received!")
22         response = PrintPlanResponse()
23         for param in request.planning_parameters:
24             if(param.parameter_name == "BedTemperature"):
25                 response.parameter_names.append(param.parameter_name)
26                 response.parameter_values.append(GetRandomBedTemp())
27
28             if(param.parameter_name == "NozzleTemperature"):
29                 response.parameter_names.append(param.parameter_name)
30                 response.parameter_values.append(GetRandomNozzleTemp())
31
32
33         print(f"Sending Plan Response Back to ARES!")
34         return response
35
36     if __name__ == "__main__":
37         planner = PrintPlanner(8006)
38         planner.DoPlanning = HandlePlanning
39         planner.start()
40
```

4.2 ANALYZERS

Analyzing is another one of the three key components to autonomous experimentation and is response for processing the output of an experiment. Much like with planners, ARES allows for the creation and use of custom analyzers that communicate over a computers local network. Keep reading to learn about creating your own analyzers.

4.2.1 ARES Analyzer Settings

The settings page in ARES contains a dedicated tab for analyzers. In this tab a list of available analyzers is displayed. By default, ARES will have two analyzers, the demo analyzer and the print analyzer. Neither of these can be edited directly from this menu, as they are protected implementations not designed to be changed. For your convenience, the print analyzer is created by default for communications with your custom build analyzer.



4.2.2 Adding New Analyzers

Educational ARES provides a static analyzer instance for you to implement your custom Python Analyzer with. This analyzer uses a simplified message scheme for communication between your analyzer instance and ARES. For this reason, no new analyzer instances should be created.

4.2.3 Creating a Custom Python Analyzer

To support your creation of a custom analyzer, a python package can be installed that handles the backend work for creating a planner. To install this package, you will need to utilize pip. Run the command “pip install PythonAres” to begin the installation process. The ARES Python package, as well as its dependencies, should be installed.

The key to building your own ARES analyzer in Python is understanding the support message structure behind it. The message structure supporting analyzer functionality is relatively simple when compared to planners. For a detailed layout of the plan request message, see section 7.

Below is a basic example of creating a new analyzer that uses port 8007. Note how the setup works very similarly to the planner. In this example, we simply return random scores between one and ten. It's important that the port you use in your Python script matches the port of the "PrintAnalyzer" in ARES. A basic template for creating one of these planners is included in your ARES installation.

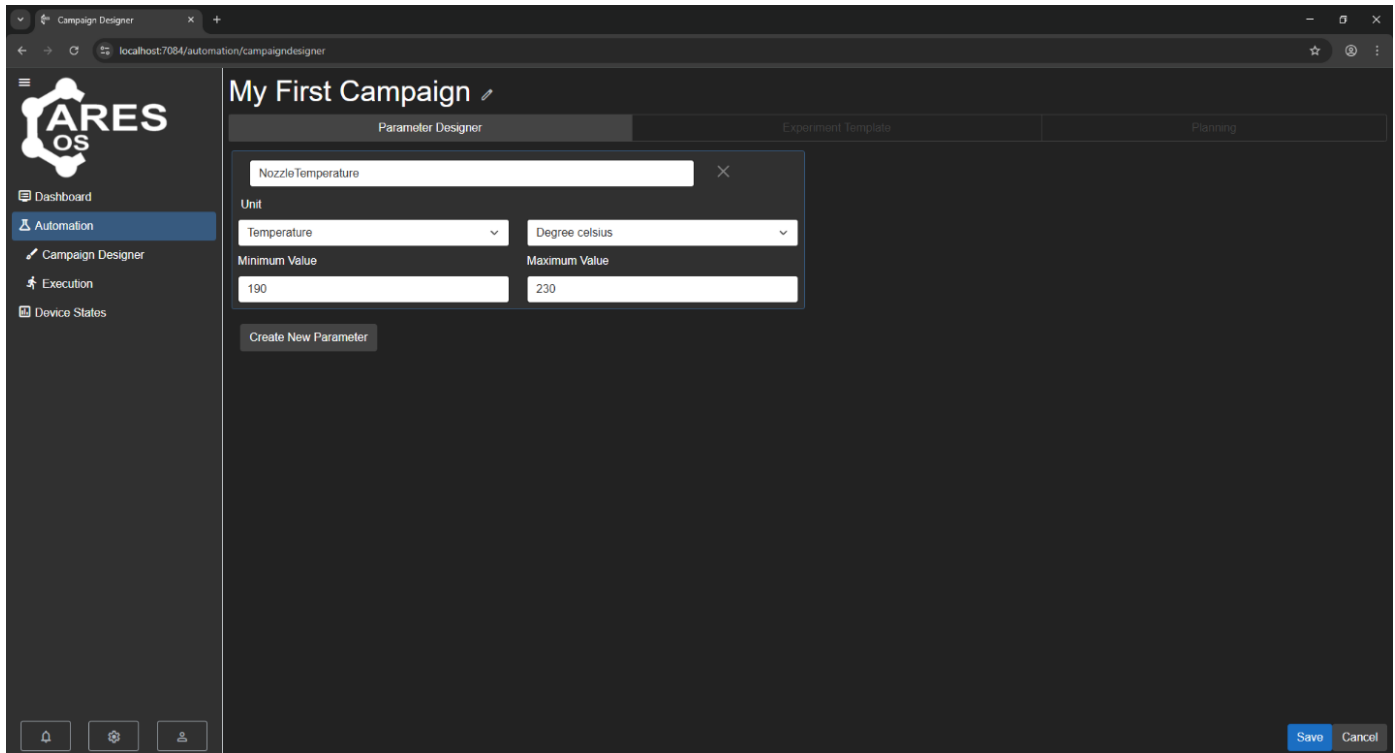
```
1  from PythonAres import PrintAnalyzer
2  from PythonAres import PrintAnalysisResponse
3  from PIL import Image
4  import random
5  import io
6
7  def AnalyzePrint(request, context):
8      response = PrintAnalysisResponse(value=random.randint(1, 10))
9      print(f"Received an ARES analysis request! Replying with {response}")
10     image = Image.open(io.BytesIO(request.image))
11     return response
12
13 if __name__ == "__main__":
14     analyzer = PrintAnalyzer(8007)
15     analyzer.UserAnalysis = AnalyzePrint
16     analyzer.start()
```

5 DESIGNING CAMPAIGNS

The construction of campaign templates is the core piece in creating an autonomous experiment. Here you will be defining the parameters and steps used in your campaign as well as your chosen methods of planning and analysis.

5.1 NAME YOUR CAMPAIGN

The first step in designing your campaign is giving your template a name. By default, new campaigns will be called "New Campaign 1", but you can edit this name by pressing the pencil icon displayed next to your template's name at the top of the screen. Enter your new campaign name, and press save to continue.



5.2 PARAMETER DESIGN

The first step in designing your campaign is defining your parameters. These are values that you will be able to plug in to your device commands when building your template later. You can define as many parameters as you need, and they require only a few pieces of information.

- Parameter Name – This will be important when creating your planner. These names will be how both ARES and your planner reference your parameters.
- Unit – The units your parameter is in, can be none.
- Minimum and Maximum Value – The constraints you place upon your parameters.

In Educational ARES there are a handful of parameters that you will typically plan for, including nozzle temperature. Below you will find an example of a filled-out parameter for nozzle temperature.

5.3 TEMPLATE DESIGN

5.3.1 Step Templates

Step templates allow you to organize your experiment into defined sections and can be given names to help identify their purpose in your experiment. To add a step template, press the “+ Add Step” button. Give the new template a name, and press create.

5.3.2 Commands

Commands are the individual actions you as the user can use on your devices. A step template can contain many commands, with each command’s construction being slightly different. Pressing the add command button brings up the command creation screen.

This screen allows you to select a device, and the command from that device you want to execute. For this example, we'll use the Prusa Printer's "move" command. When selected a brief description of the command appears, along with all the inputs the command requires. In our example, we have four inputs: X, Y, Z and Dwell Time. Enter your desired values for these parameters. In this example, the printer will move to the coordinates 100,100,100 and dwell there for 10 seconds. Note that dwell time can be zero if none is desired. Press 'create' to finalize your command and add it to your experiment.

Command Create

Device

Prusa Printer

Command

Move

A command that tells the print head to move to a specific location with an optional dwell time value. If no dwell is desired, enter zero.

X - Coordinate

100

Y - Coordinate

100

Z - Coordinate

100

DwellTime - Seconds

10

☐ Provides Experiment Output

Create

Cancel

5.3.3 Planned Parameters

Using your planned parameters is made straight forward by ARES. To demonstrate this, lets add a “Print” command for your Prusa Printer. You can see the print command has several parameters. The first is your G-Code file. It’s important to note that you must upload your file as G-Code and **not** binary G-Code. You can either drag and drop the file into the box or press the button to browse your computers files. A successfully uploaded file will display the name of your file, as well as its size.

Command Create

Device

Prusa Printer

Command

Print

A command that tells the printer to print with the provided G-Code source as a base file. Includes the ability to modify several parameters of that G-Code such as nozzle temperature, extrusion rate and acceleration.

Drag and drop or click to choose files

NozzleTemperature - Degree's Celsius

☐ Planned

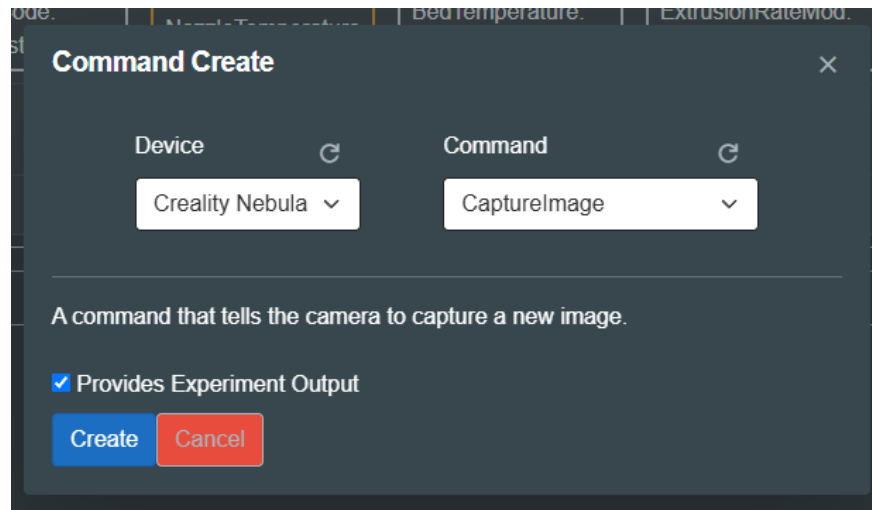
BedTemperature - Degree's Celsius

Next, let's add a planned parameter. Under certain parameter there is a check box labeled "Planned". Checking this will give you a drop-down list of all available parameters in your experiment template. Select the parameter we created earlier called "Nozzle Temperature". Assigning this to your nozzle temperature parameter on your print command ensures that whatever value your planner chooses for nozzle temperature is sent to the printer.

You can now press save to create your command. All planned parameters will appear with a yellow border on the experiment template page. The print command only requires a provided G-Code file. Any parameters not given a value will be assumed to be a default value, and those properties will not be changed during your print. Not all parameters are plannable. Parameters that **cannot** be planned for won't have a "Planned" checkbox and will have to be given manual values.

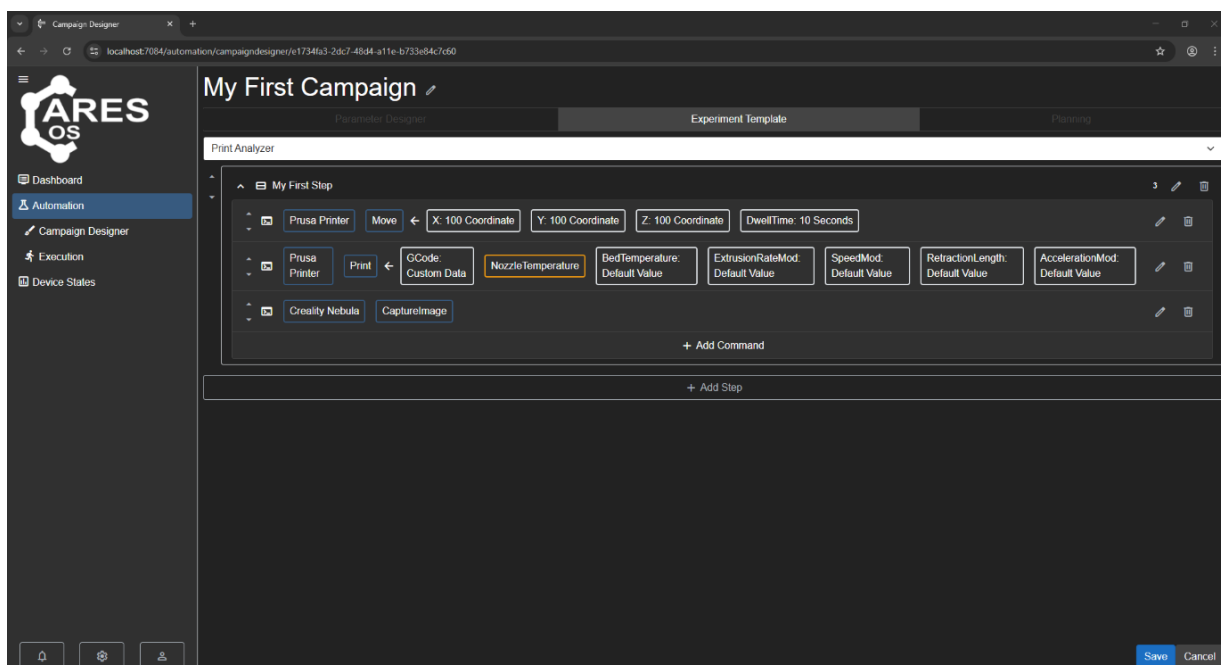
5.3.4 Experiment Output

To provide your analyzer with meaningful output, you need a command that is marked to provide output for our experiment. To accomplish this, you can add a command from your camera device to capture an image. Add a command and select the name of your camera device. For the command, pick “CaptureImage”. The capture image command is very basic and takes no input parameters. To send the output of this command to your analyzer, check the “Provides Experiment Output” box as shown below. Press the create button, you have successfully added output to your experiment.



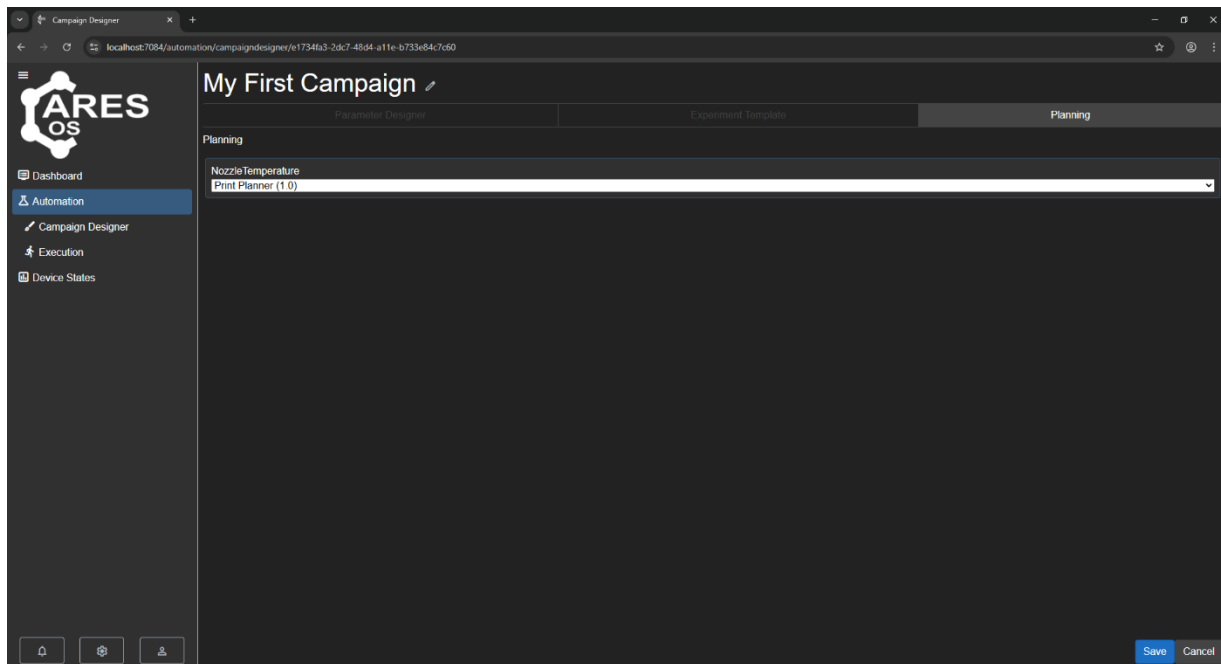
5.3.5 Analyzer Assignment

The last item you need to change in your experiment template is your analyzer selection. Above your steps, there is a drop-down menu that by default reads “Select Analyzer”. Clicking this will display all the analyzers currently available for use. For now, you should select the “Print Analyzer”, as this is the custom analyzer you’ve written in Python. With this completed you can move on to the “Planning” tab of your experiment designer!



5.4 PLANNING ASSIGNMENT

The last step to designing your campaign is to assign planners to generate values for your variables. In your example, we only have one parameter named “NozzleTemperature”, but any number of parameters you create will appear on this screen. Pressing the “Select a planner” drop-down will display a list of all available planners. Select the “Print Planner” to assign the planning of this variable to your custom Python planner.



5.5 OVERVIEW

You can now press the save button in the bottom right corner of your screen. ARES will save your experiment template to your local ARES data directory. If you ever want to edit this template, you can do so by returning to the campaign designer tab and pressing the edit button on your campaign template. Campaign templates can also be downloaded from the campaign designer tab and are saved in a JSON data format. Now that you have built your campaign template, it's time to run your first experiment.

6 EXECUTION

6.1 TEMPLATE SELECTION

Navigate to the execution tab located under the autonomy section. The first step to executing your experiment is selecting a template. Using the drop-down menu, select the experiment template you just finished building. ARES will populate a preview of the experiment, showing the steps and commands, you have created.

6.2 STOP CONDITION

To start experimenting, ARES needs to know when to stop! This can be set in a few ways. ARES allows you to either set the number of experiments to run, or to stop when a desired experiment result is achieved within a defined margin of error. Educational ARES offers a Smart Print mode, which you can read more about below.

6.2.1 Smart Print Calculation

A unique feature to Educational ARES is the ability to ask ARES to calculate a set number of experiments that can be run printing your object repeatedly on a single print bed. This means ARES will determine the size of the item you provided G-Code for and attempt to print as many of them as it can. To use the smart print mode, simply press the toggle switch. ARES will populate the desired number of experiments box and lock it from being changed until smart print is disabled.

6.2.2 Not Utilizing Smart Print

Using the smart print feature can make the experimentation process an entirely hands off process, but there are reasons you might choose to instead do a set number of experiments without smart print turned on. ARES supports this functionality, but there are some important considerations to make before using it. If you choose not to enable smart print, you can set any number of experiments to be run, but ARES will not alter the positioning of the object being printed. **The print bed will need to be cleared before the next print starts.** You can add sleep commands to your experiment template if you would like to run an experiment in this way and remove the objects yourself in between experiments.

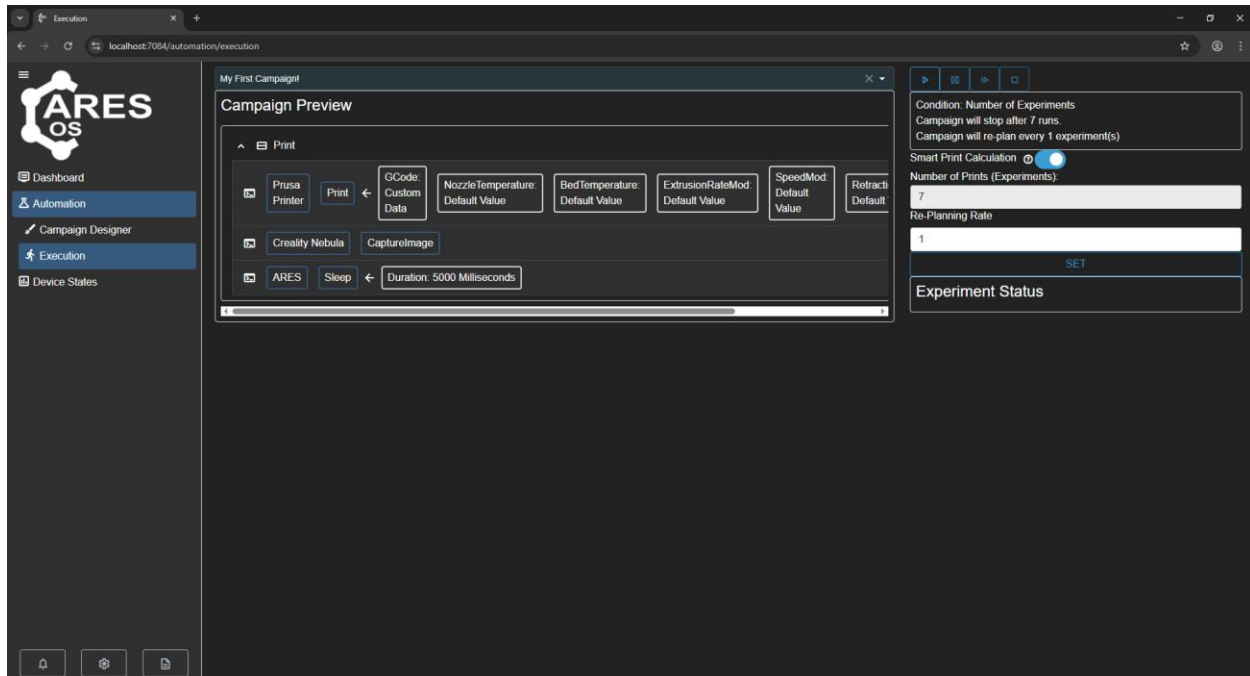
To set the number of experiments you want to run manually, enter how many experiments you want to complete in the “Desired Number of Experiments” box and then press set. If this value is taken, ARES will populate a box with execution information above this.

6.3 RE-PLANNING RATE

ARES gives you the ability to specify how often you want your planner to generate new values for your experiment parameters, this is called the re-planning rate. By default, ARES will set the re-planning rate to one. This means that every experiment ARES will ask your planner to generate new values for all planned experiment parameters. If this is your desired behavior, then no change is required, and you can move on to the next step. Otherwise enter the number of experiments you wish to run before regenerating your parameter values, then press the set button.

6.4 STARTING YOUR CAMPAIGN

With your template selected, re-planning rate and stop condition set you are ready to start your first campaign. ARES will display information about your experiments execution at the top right of the screen, including your stop condition and re-plan rate. Check these values and correct them if they don't match what you would expect. If no information has appeared as shown below, ensure you have hit set on your stop condition. Ensure your planner and analyzer are running prior to starting your campaign. When you are satisfied with your campaign setup, press the play button to begin experimenting.



6.5 CAMPAIGN EXECUTION

ARES will display a notification when the campaign successfully starts. Once experimentation starts, ARES will display its status on the execution page. You can stop the experiment by pressing the stop button at the top right of the execution page. When the stop button is pressed, ARES will stop the experiment when the current command finishes. The campaign will automatically finish based on the earlier set stop condition.

7 MESSAGE STRUCTURE

7.1 PLANREQUEST

The plan request message is used by ARES to request new planning values from an associated planner.

- `planning_parameters` (repeated `PlanningParameter`, field number 1)
 - A list of the available planning parameters ARES has provided the planner.

7.2 PLANNINGPARAMETER

A parameter provided to planners with supporting information from ARES.

- `parameter_name` (string, field number 1)
 - The name of the parameter.
- `parameter_value` (double, field number 2)
 - The value of the parameter.
- `minimum_value` (double, field number 3)
 - The minimum allowed value of the parameter.
- `maximum_value` (double, field number 4)
 - The maximum allowed value of the parameter.
- `minimum_precision` (double, field number 5)
 - The precision of the given parameter.
- `parameter_history` (repeated double, field number 6)
 - A list of historical values given to the parameter.
- `data_type` (string, field number 7)
 - The corresponding type of the parameter.
- `is_planned` (bool, field number 9)
 - A bool value to determine whether a parameter is designated to be planned for.
- `is_result` (bool, field number 10)
 - A bool value to determine whether a parameter is designated at the output value of the experiment.

7.3 PLANRESPONSE

The plan response message is used by planners to send new planned parameters back to ARES.

- `parameter_names` (repeated string, field number 1)
 - A list of parameter names that have been planned for.
- `parameter_values` (repeated double, field number 2)
 - A list of parameter values that have been planned for.

7.4 ANALYZEPRINT

The analyze print message serves as the analysis request message that ARES sends to corresponding analyzers.

- image (bytes, field number 1)
 - An array of bytes that represent the image data being captured by ARES.

7.5 PRINTANALYSISRESPONSE

The print analysis response message is designed to be used by external planners to send the result of analysis back to ARES.

- value (double, field number 1)
 - A double representing the given score the analyzer has given the latest image.